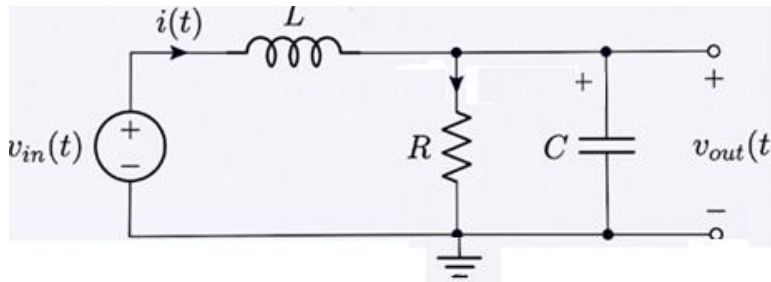


A second -order electrical circuit is shown in the figure below, where v_{in} is the input and v_{out} is the output. $L = 1$ H, $C = 1$ F. Do the problem for $R = 1 \Omega$ and $R = 0.25 \Omega$



1. Find the network function of the circuit.
2. Determine the poles of the network function.
3. Find the state variable representation of the circuit. Take $i(t)$ as x_1 (with the current direction shown in the figure) and the voltage across the capacitance as x_2 (with the polarity shown).
4. Determine the eigenvalues of the A matrix
5. Obtain the network function from the state variable representation.
6. What is the natural frequency and damping ratio when $R = 1 \Omega$ and $R = 0.25 \Omega$
7. Using the network function approach, find the unit step response of the circuit. $R = 1 \Omega$
8. Using the state variable representation, assuming the initial conditions are zero, determine unit step response of the circuit.
9. Find the equilibrium point from the state equations when $R = 1$
10. With the help of the figure, justify the answer obtained in question 9.
11. Solve the state equations using the MATLAB command `ode23` and plot the time variation of the state variables when $R = 1 \Omega$ and $R = 0.25 \Omega$. Also, plot the state trajectories for both the cases.
12. Examine the nature of equilibrium points.

Extra question (No marks): What is the difference in the output (unit step response) when $R = 1 \Omega$ and $R = 0.25 \Omega$